# Smart Sandbag System

Comprehensive Technical Report & Developer Guide v3.0

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**Abstract**

This document serves as the comprehensive technical foundation and operational manual for the Smart Sandbag System. It details the transition from a localized prototype to a fully cloud-hosted, IoT-enabled sports platform. Key expansions in this version include the mathematics of the dynamic physics engine, the Progressive Web App (PWA) deployment mechanics, cross-platform Web Bluetooth (BLE) handling, gamification subsystems, and extensive end-user workflows including data export for Large Language Model (LLM) analysis.

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# 1 Introduction

The Smart Sandbag System bridges the gap between traditional combat sports training and modern IoT analytics. By utilizing a Seeed Studio XIAO edge-computing microcontroller embedded within a heavy bag, the system captures high-fidelity kinetic data via an accelerometer and localizes impact sites using a 6-microphone condenser array.

Recent updates have transitioned the platform from a local developer environment to a continuous 24/7 cloud server. This enables a "many-users-to-one-bag" paradigm, fundamentally undercutting the cost of individual smart boxing gloves while offering superior data aggregation, real-time audio coaching, and scalable gamification.

# 2 Physics Engine & Dynamic Force Derivation

The core function of the system is translating raw linear acceleration ($a$) measured in G-forces into rotational impact force ($F$) measured in Newtons. The system models the heavy bag as a physical pendulum swinging from a fixed ceiling pivot.

## 2.1 System Constants and User Calibration

To ensure accuracy across different gyms and equipment, the physics engine relies on variables stored securely in the Django backend:

- $m_{bag}$: Mass of the sandbag (kg)
- $L_{bag}$: Length of the sandbag (m)
- $L_{chain}$: Length of the mounting chain (m)

## 2.2 Moment of Inertia ($I$)

Using the parallel axis theorem, the bag is treated as a uniform solid cylinder rotating around a pivot point located at a distance $L_{chain}$ from the top of the cylinder. The Moment of Inertia is calculated as:

$$I = m_{bag} \times \left( \frac{L_{bag}^2}{12} + \left( L_{chain} + \frac{L_{bag}}{2} \right)^2 \right) \tag{1}$$

## 2.3 Dynamic Impact Radius ($r_{impact}$)

With the 6-microphone array, the system dynamically calculates the exact distance from the pivot point to the impact site. The bag is divided into 7 equal segments ($L_{bag}/7$). The specific hit distance from the top of the bag ($Y_{hit}$) updates the impact radius dynamically per punch:

$$r_{impact} = L_{chain} + Y_{hit} \tag{2}$$

## 2.4 Force Derivation ($F$)

Assuming the IMU sensor is housed at the **bottom** of the bag to improve the signal-to-noise ratio and protect the board from low strikes, the angular acceleration ($\alpha$) is derived as:

$$\alpha = \frac{a}{L_{chain} + L_{bag}} \tag{3}$$

Because Torque ($\tau$) is defined as both $\tau = I \times \alpha$ and $\tau = F \times r_{impact}$, we isolate Force ($F$):

$$F = \frac{I \times a}{r_{impact} \times (L_{chain} + L_{bag})} \tag{4}$$

# 3 Hardware & Firmware Optimizations

To map shot distribution effectively while preserving Bluetooth bandwidth, the hardware and software must operate in tight synchronization.

## 3.1 Hardware Envelope Detection & Digital Noise Gate

To prevent the microcontroller from sampling the 0V trough of an audio wave during a 50ms polling cycle, each microphone is equipped with an Envelope Detector (diode + capacitor + bleed resistor).

Furthermore, to suppress "ghost punches" generated by floating analog pins acting as electromagnetic antennas, a digital Noise Gate is hardcoded into the C++ firmware. Assuming a 10-bit ADC (range 0–1023), any analog reading below 150 is forcefully suppressed to zero.

## 3.2 Bluetooth Payload Optimization

Standard BLE MTU (Maximum Transmission Unit) sizes strictly limit payload length. Sending six individual microphone traits and three IMU traits simultaneously overflows the buffer. To solve this, the firmware packs the 6 peak microphone values into a single contiguous array of 16-bit integers (`int16_t`), requiring exactly 12 bytes. This is transmitted over a single characteristic and unpacked on the client side using Little-Endian byte order.

# 4 Cloud Architecture & Software Subsystems

The application has been migrated from local execution to a continuous cloud server, establishing a stable, 24/7 service architecture.

## 4.1 Progressive Web App (PWA) Implementation

The web portal acts as a downloadable native application. This is achieved via two core components integrated into the static file directory:

1. **manifest.json:** Defines the application metadata, theme colors (#007bff), and home-screen icon assets (192x192 and 512x512 PNGs).
2. **Service Worker (sw.js):** A background script that registers the application state and caches assets for faster loading, prompting the "Add to Home Screen" browser banner.

## 4.2 Cross-Platform Web Bluetooth Constraints

The system utilizes the Web Bluetooth API to bypass traditional app store deployment. However, hardware manufacturers enforce strict security constraints:

- **Android Devices:** Fully supported via the Google Chrome browser. Users must ensure both Bluetooth and Location Services (GPS) are enabled for the BLE scanner to operate.
- **iOS Devices (Apple):** Apple strictly blocks the Web Bluetooth API within Safari and all iOS browsers. To connect to the sandbag, iOS users *must* use a specialized IoT browser application, such as **Bluefy**.

### 4.3 Gamification & Audio Feedback

To enhance user retention and provide immediate tactical feedback, two new subsystems have been introduced:

- **Web Speech API:** A JavaScript listener triggers upon receiving payload data, audibly announcing the derived Force (e.g., "850 Newtons!"). This allows the user to maintain visual focus on the target.
- **Achievement Badges:** The Django backend evaluates the user's `Session` and `Punch` aggregates. Visual badges are unlocked and displayed dynamically when users hit specific milestones: 10, 100, and 1,000 total punches, or recording a single strike exceeding 1,000 N.

### 4.4 Data Export & LLM Integration

Coaches and athletes require raw data for longitudinal performance tracking. The system now features a dedicated Django view that compiles session objects into a structured `.csv` file. This file includes Punch Number, Force, Target Zone, and exact Timestamps. Users can download this file and input it directly into Large Language Models (e.g., ChatGPT) to generate customized training regimens and endurance reports.

## 5 Developer Guide: Environment Setup

### 5.1 Local Development (Testing)

To deploy the local Django development environment for testing and feature development:

```
# 1. Clone the repository
git clone https://github.com/ZuonVu/Smart-Sandbag.git
cd Smart-Sandbag

# 2. Create and activate a virtual environment
python -m venv venv
# On Windows: .\venv\Scripts\activate
# On Mac/Linux: source venv/bin/activate

# 3. Install dependencies and migrate database
pip install -r requirements.txt
python manage.py migrate

# 4. Start the local server
python manage.py runserver
```

### 5.2 Microcontroller Flashing (Arduino IDE)

To upload the `XIAO_code.ino` firmware:

1. Open Arduino IDE. Navigate to **Sketch > Include Library > Manage Libraries**.
2. Install **ArduinoBLE** (by Arduino) and **LSM6DS3** (by Seeed Studio).
3. Select the correct XIAO board and COM port, then click **Upload**.

## 6 End-User Manual & Demonstration Guide

The Smart Sandbag System is designed for zero-friction user onboarding. No terminal commands or installations are required by the end-user.

### 6.1 Accessing the System (Demonstration Day)

For public demonstrations, users can access the system quickly via QR code:

1. Scan the provided QR code placed near the heavy bag.
2. If using an Android, open in Chrome. If using an iPhone, open in Bluefy.
3. Log in using the public demonstration account:
   - **Username:** Coach
   - **Password:** Coachforboxing

### 6.2 System Calibration & Connection

1. Navigate to the **Settings** tab and verify that the Bag Mass, Bag Length, and Chain Length match the physical equipment in the gym.
2. Stand near the sandbag and click **Connect Sandbag** on the dashboard.
3. A browser prompt will appear. Select the device named **"Sandbag"** and click Pair.

### 6.3 Executing a Training Session

1. Click **Start Session** to open the database transaction.
2. Strike the bag. The system will automatically filter out light taps under 2.5G.
3. Listen to the live audio feedback and watch the UI map your shot distribution across the 6 target zones.
4. Unlock achievement badges as you hit endurance and power milestones.
5. When finished, click **End Session** to save your data permanently.

### 6.4 Reviewing History and Exporting Data

Navigate to the **History** tab to view past sessions. The dashboard provides aggregate statistics such as your hardest punch and favorite target zone. To analyze your performance further, click the **Download CSV for Coach** button. This file can be shared with your trainer or uploaded to an AI tool to receive personalized feedback. Test sessions can be safely removed using the Delete button.